Exercise 1 :

$$Y_t = e_t + c \sum_{k=1}^{t} e_{t-k}$$

a) by direct computation :

$$E[Y_t] = \underbrace{E[e_t]}_{0} + c \sum_{k=1}^{t} \underbrace{E[e_{t-k}]}_{0}$$

$$\boxed{E[Y_{t-k}] = 0}$$

$\forall\ k \in \mathbb{N}$,

$$\begin{aligned} Cov(Y_t, Y_{t-k}) &= Cov\left(e_t + c\sum_{i=1}^{t} e_{t-i},\ e_{t-k} + c\sum_{j=1}^{t-k} e_{t-k-j}\right) \\ &= Cov(e_t, e_{t-k}) + c\sum_{j=1}^{t-k} Cov(e_t, e_{t-k-j}) \\ &\quad + c^2 \sum_{i=1}^{t}\sum_{j=1}^{t-k} Cov(e_{t-i}, e_{t-j}) \\ &\quad + c\sum_{i=1}^{t} Cov(e_{t-i}, e_{t-k}) \\ &= \begin{cases} \sigma_e^2(1 + c^2 t) & \text{, if } k = 0 \\ c(1 + c^2(t-k))\sigma_e^2, & \text{if } 1 \le k \le t \\ 0 \text{ if } & k > t \end{cases} \end{aligned}$$

$Cov(Y_t, Y_{t-k})$ is a function of t so (Y_t) is not stationary.

b)

$$\nabla y_t = y_t - y_{t-1} = e_t - e_{t-1} + c\sum_{k=1}^{t} e_{t-k} - c\sum_{k=1}^{t-1} e_{t-k-1}$$

$$= e_t + e_{t-1}(c-1)$$

$$E[\nabla y_t] = \underbrace{E[e_t]}_{0} + \underbrace{E[e_{t-1}]}_{0}(c-1)$$

$$E[\nabla y_t] = 0$$

$\forall k \in \mathbb{N}$,

$$\text{cov}(\nabla y_t, \nabla y_{t-k}) = \text{cov}(e_t + (c-1)e_{t-1},\ e_{t-k} + (c-1)e_{t-k-1})$$

$$= \text{cov}(e_t, e_{t-k}) + (c-1)\,\text{cov}(e_{t-1}, e_{t-k}) + (c-1)\,\text{cov}(e_t, e_{t-k-1}) + (c-1)^2\,\text{cov}(e_{t-1}, e_{t-k-1})$$

$$\gamma_k = \begin{cases} \sigma_e^2(1+(c-1)^2) & \text{, if } k = 0 \\ (c-1)\sigma_e^2 & \text{, if } k = 1 \\ 0 & \text{, o.w} \end{cases}$$

$\{\nabla y_t\}$ is stationary because the mean is a constant function of t and γ_k only depends on the lag k.

b)

$d = 1$ because one differenciation is enough to have stationarity

$p = 0$ no autoregression

$q = 1$ only one past error term is used

conclusion: $\{y_t\}$ is an ARIMA(0, 1, 1) process.